Form or Substance? Institutional Pressures and Accountability in Public-Sector Financial Reforms

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Abstract

Purpose. Governments mandate accrual accounting and performance-based budgeting, yet adoption on paper often fails to change decisions. This study asks how coercive, normative and mimetic pressures push executive agencies towards substantive implementation or symbolic compliance, how these responses affect financial accountability through the usefulness of reported information, and whether the maturity of the financial information system conditions the process. **Design/methodology/approach.** A survey of 336 financial managers, comptrollers, budget experts and accountants in executive agencies was analysed with covariance-based structural equation modelling in which every item loads only on its own latent variable (no composites), latent interaction terms, and 2,000-draw bootstrap estimates of conditional indirect effects. **Findings.** Coercive pressure raised both substantive implementation ($\beta = 0.32$) and symbolic compliance ($\beta = 0.38$); normative pressure raised the former ($\beta = 0.37$) and lowered the latter ($\beta = -0.22$); mimetic pressure raised symbolic compliance only ($\beta = 0.33$). Substantive implementation increased, and symbolic compliance reduced, both the usefulness of information and accountability, and usefulness mediated part of these effects. Maturity amplified the usefulness payoff of substantive implementation and dampened the translation of coercion into symbolic compliance. The net effect of coercion was close to zero because its constructive and ceremonial routes offset one another, while normative pressure had a clear positive net effect. **Originality/value.** The study separates substantive from symbolic responses to the same pressures and shows when mandates translate into accountability. **Practical implications.** Reform agencies should supplement mandates with professional norms, monitor use rather than submission of reports, and invest in integrated, auditable systems.

Keywords: institutional theory; decoupling; accrual accounting; performance-based budgeting; public-sector accountability; information systems

JEL classification: M41; M48; H83; D73

Paper type: Research paper

1. Introduction

Over the past four decades, public-sector accounting and budgeting have been reorganised around the idea that governments should manage by information on costs, assets, liabilities and results rather than by cash and appropriations (Hood, 1991; Lapsley, 2009). Accrual accounting, internationally promoted through the International Public Sector Accounting Standards, and performance-based budgeting, which links appropriations to outputs and outcomes, are the two flagship instruments of this agenda (Pollitt & Bouckaert, 2017; Christiaens et al., 2015). Their promise is better decisions and stronger accountability to legislatures, auditors and citizens.

The evidence on delivery is mixed. Cross-country studies show that countries adopt the same standards in markedly different ways (Pina et al., 2009; Christiaens & Reyniers, 2010), and the literature on performance management documents that producing performance information is not the same as using it (Moynihan, 2008; Moynihan & Pandey, 2010; Kroll,

2015). Iran illustrates the gap. Public-sector accounting standards and performance-based budgeting have been introduced through legislation and development plans, but recent Iranian studies report partial implementation, legal and structural obstacles, and weak use of information: Ahmadi et al. (2023) estimate that performance-based budgeting is implemented at roughly 34, 26 and 31 per cent in strategic planning, cost analysis and performance management respectively, and Nourani and Akhlaghi Yazdinejad (2023) identify legal, structural and environmental bottlenecks to implementing public-sector accounting standards. Qualitative models (Pourghaffar et al., 2022; Naroui et al., 2020) map what a good system should contain, but we know little about what drives agencies to implement reforms substantively rather than ceremonially, and what each response costs or yields in accountability.

Institutional theory offers a precise language for this puzzle. Organisations facing coercive, normative and mimetic pressures adopt structures to gain legitimacy (DiMaggio & Powell, 1983), and formal adoption may be decoupled from operations (Meyer & Rowan, 1977; Bromley & Powell, 2012). Yet three gaps remain. First, empirical accounting studies rarely treat substantive implementation and symbolic compliance as two distinct outcomes of the same pressures; they tend to measure "adoption" as a single construct. Second, the mechanism linking implementation to accountability—whether reported information is actually useful to those who decide—has seldom been modelled. Third, the role of the information infrastructure in making ceremonial behaviour visible, and substantive behaviour productive, is under-theorised, despite the emphasis on integrated digital systems in public-management reform (Dunleavy et al., 2006; DeLone & McLean, 2003).

This paper addresses these gaps with a survey of financial and budget staff in Iranian executive agencies. It makes three contributions. (i) It shows that institutional pressure is not a single cause with a single effect: coercion drives both substantive and symbolic responses, normative pressure favours substance, and mimetic pressure breeds form. (ii) It identifies usefulness of reported information as the bridge from implementation to accountability and quantifies the offsetting routes that make the net effect of coercion small. (iii) It demonstrates that information-system maturity conditions both the payoff of substantive implementation and the pull of coercion towards ceremony. Methodologically, the study estimates a fully latent model with latent interactions and bootstrap conditional indirect effects instead of the summed or averaged scale scores that dominate the field.

The remainder of the paper is organised as follows. Section 2 develops the theory and hypotheses; Section 3 describes data, measures and methods; Section 4 reports results, including competing models and robustness checks; Section 5 discusses theoretical, practical and policy implications; Section 6 outlines limitations and future research. Appendices report the full instrument, measurement and structural estimates, and supplementary analyses.

2. Theory and hypotheses

2.1 Institutional pressures and organisational responses

DiMaggio and Powell (1983) distinguish three mechanisms of institutional isomorphism. Coercive pressure stems from formal and informal pressures exerted by organisations on which an agency depends, and from legal mandates; normative pressure arises from

professionalisation, training and occupational networks; mimetic pressure arises when uncertainty leads organisations to imitate peers perceived as legitimate. Scott (2014) links these mechanisms to the regulative, normative and cultural-cognitive pillars of institutions. Empirically the three are correlated but not interchangeable, and research on their effects shows that their consequences differ (Mizruchi & Fein, 1999; Teo et al., 2003).

Organisations are not passive recipients. Oliver (1991) describes a repertoire of strategic responses from acquiescence to avoidance, and Kostova and Roth (2002) separate ceremonial adoption—the practice is adopted formally without understanding or commitment—from internalisation, in which employees value and use it. In public accounting, performance measures and accrual statements are particularly prone to ceremonial use because they are visible to superiors, legible as signals of modernisation, and costly to verify (Brignall & Modell, 2000; Modell, 2009). We therefore define two organisational responses. **Substantive implementation** is the use of programme structures, cost information and accrual statements in preparing budgets and in managerial decisions. **Symbolic compliance** is the production of reports and indicators mainly for submission to higher authorities, with indicators that are easy to meet and little effect on resource allocation.

2.2 Pressures and responses (H1–H5)

Coercive pressure makes non-adoption costly and behaviour observable. Where legal mandates and oversight are strong, agencies have reasons both to implement and to display implementation (DiMaggio & Powell, 1983; Oliver, 1991). Because mandates do not distinguish between the two, we expect coercion to raise both responses (H1, H2). Normative pressure operates by shaping professional identity: accountants and comptrollers who see accrual information as a mark of professional competence internalise it, which should raise substantive implementation and reduce reliance on ritual (H3, H4) (DiMaggio & Powell, 1983; Greenwood et al., 2011). Mimetic pressure, by contrast, supplies templates; imitation is cheap, copies form rather than function, and is most attractive when the agency lacks the capacity or incentive to internalise. We therefore expect mimetic pressure to raise symbolic compliance (H5) but not to promote substantive implementation, a path we estimate but do not hypothesise.

H1: Coercive pressure is positively related to substantive implementation. **H2:** Coercive pressure is positively related to symbolic compliance. **H3:** Normative pressure is positively related to substantive implementation. **H4:** Normative pressure is negatively related to symbolic compliance. **H5:** Mimetic pressure is positively related to symbolic compliance.

2.3 From implementation to usefulness and accountability (H6–H10)

Reforms create value only when information is used (Moynihan, 2008; Moynihan & Pandey, 2010). When agencies implement substantively, cost and accrual information enters budget preparation and management meetings and is therefore timely, understandable and relevant; when compliance is symbolic, reports are produced for form, are less trusted internally and are discounted by users (Kroll, 2015; Modell, 2009). We define the usefulness of reported information as the extent to which information is the basis of managerial decisions, is timely and understandable, gives a fuller picture than cash accounting and is reliable for evaluating efficiency and effectiveness. Accountability, following Bovens (2007),

is a relationship in which the actor informs, justifies and accepts consequences; it is also a core input to public value (Moore, 1995; Bracci et al., 2015). Substantive implementation should strengthen accountability directly, by making deviations traceable and responsibilities clear, and indirectly, through usefulness; symbolic compliance should weaken it by providing information that cannot be verified or acted upon.

H6: Substantive implementation is positively related to usefulness of information. **H7:** Symbolic compliance is negatively related to usefulness of information. **H8:** Substantive implementation is positively related to financial accountability. **H9:** Symbolic compliance is negatively related to financial accountability. **H10:** Usefulness of information is positively related to financial accountability.

2.4 The conditioning role of information-system maturity (H11–H12)

Information-systems research holds that the benefits of information depend on system and information quality (DeLone & McLean, 2003), and digital-era governance stresses integration, reintegration of functions and transparency (Dunleavy et al., 2006). We define maturity as the integration, real-time availability and auditability of the agency's financial systems. Maturity should play two roles. First, integrated and timely data allow substantive implementation to deliver usable information; maturity should therefore strengthen the effect of substantive implementation on usefulness (H11). Second, auditable trails and traceable changes raise the cost of reporting that is detached from practice; maturity should therefore weaken the pathway from coercive pressure to symbolic compliance, because mandated reporting becomes verifiable rather than ceremonial (H12).

H11: Information-system maturity strengthens the positive effect of substantive implementation on usefulness of information. **H12:** Information-system maturity weakens the positive effect of coercive pressure on symbolic compliance. **H13 (mediation):** Institutional pressures exert indirect effects on accountability through substantive implementation and symbolic compliance, via usefulness.

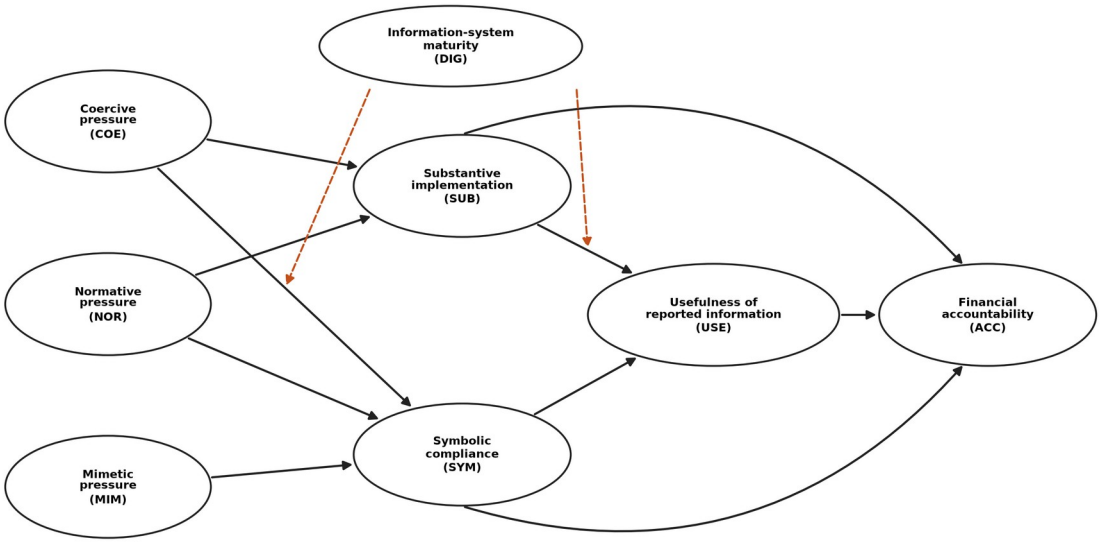


Figure 1. Conceptual model. Solid arrows: structural paths; dashed orange arrows: moderation by information-system maturity.

3. Data and method

3.1 Context, sample and procedure

The study population comprises financial managers, comptrollers, budget experts and accountants or internal auditors of executive agencies. The analysed sample has 336 respondents: 35% from ministries and national organisations, 30% from universities and health-care institutions, 18% from municipalities and local bodies and 17% from provincial offices; the mean professional experience is 13.8 years (SD = 7.0) and the mean age 45.3 years (SD = 7.3). [Sampling procedure, fieldwork period, distribution channel and response rate must be completed from the actual administration.] Respondents rated the agency in which they work; anonymity was assured and answers are analysed in aggregate.

With 116 free parameters the ratio of observations to parameters is 2.9:1, below the conventional 10:1 rule of thumb. We therefore base the adequacy of the sample on power analysis for covariance-structure models (MacCallum et al., 1996): for the degrees of freedom of the structural model, power to reject close fit (RMSEA = .05) against RMSEA = .08 exceeds .99. Because latent variables with multiple reflective indicators absorb measurement error, and the loadings are high, we regard N = 336 as adequate for the model, but we report DWLS and trimmed-sample estimates as checks.

3.2 Measures

All constructs are measured with five-point agreement items (Appendix A). Table 1 shows the origin of each scale. The three pressure scales are adapted from the institutional-pressure measures of Teo et al. (2003), re-worded for budgeting and accrual reforms; substantive implementation, symbolic compliance and accountability are researcher-developed from the literature; maturity and usefulness combine adapted and new items. No scale is a published, validated instrument in this exact form, so we provide extensive validity

evidence below. Content validity was assessed with the content validity ratio and index (Lawshe, 1975; Polit & Beck, 2006) [report the number of experts and the actual CVR/CVI values]. Latent variables are never summed or averaged: every item is an indicator of its own latent variable only.

Table 1. Constructs, number of items, origin of the scales and principal sources

Construct	Items	Origin	Principal sources
Coercive pressure (COE)	4	Adapted	DiMaggio & Powell (1983); Teo et al. (2003)
Normative pressure (NOR)	4	Adapted	DiMaggio & Powell (1983); Teo et al. (2003)
Mimetic pressure (MIM)	4	Adapted	DiMaggio & Powell (1983); Oliver (1991); Teo et al. (2003)
Substantive implementation (SUB)	5	Researcher-developed	Pollitt & Bouckaert (2017); Moynihan (2008); Christiaens et al. (2015)
Symbolic compliance (SYM)	4	Researcher-developed	Meyer & Rowan (1977); Bromley & Powell (2012); Modell (2009)
Information-system maturity (DIG)	4	Adapted + new	DeLone & McLean (2003); Dunleavy et al. (2006)
Usefulness of reported information (USE)	4	Adapted + new	Moynihan & Pandey (2010); Kroll (2015); DeLone & McLean (2003)
Financial accountability (ACC)	5	Researcher-developed	Bovens (2007); Bracci et al. (2015); Moore (1995)

3.3 Estimation strategy

Models are estimated by maximum likelihood in semopy (Igolkina & Meshcheryakov, 2020). Fit is judged jointly: CFI and TLI ≥ .95, RMSEA ≤ .06 and SRMR ≤ .08 (Hu & Bentler, 1999); the χ^2 test is reported but, as is well known, over-rejects in samples of this size. Convergent validity uses standardized loadings, composite reliability (CR) and average variance extracted (AVE) (Fornell & Larcker, 1981); discriminant validity uses the Fornell–Larcker criterion and the heterotrait–monotrait ratio (Henseler et al., 2015). Common-method bias is probed with Harman's single-factor test, a one-factor CFA and a common-latent-factor model (Podsakoff et al., 2003). Two latent interaction terms (maturity × substantive implementation; maturity × coercive pressure) are specified with matched-pair product indicators (Marsh et al., 2004) and, as a robustness check, residual-centred indicators (Little et al., 2006). Indirect and conditional indirect effects are evaluated at −1, 0 and +1 SD of maturity with 2,000 nonparametric bootstrap draws and percentile intervals (Preacher & Hayes, 2008); indices of moderated mediation follow Hayes (2015). Latent R^2 equals one minus the standardized residual variance, and the effect size of an interaction is f^2 from the change in R^2. Five competing structures are compared by chi-square difference tests and fit indices.

4. Results

4.1 Descriptive checks

Item skewness ranged from −0.62 to 0.40 and excess kurtosis from −0.81 to −0.14, well within common tolerances. Mardia's multivariate kurtosis (1211 versus 1224 expected) indicated slight departure from multivariate normality, which motivates the DWLS check in Section 4.6.

4.2 Measurement model

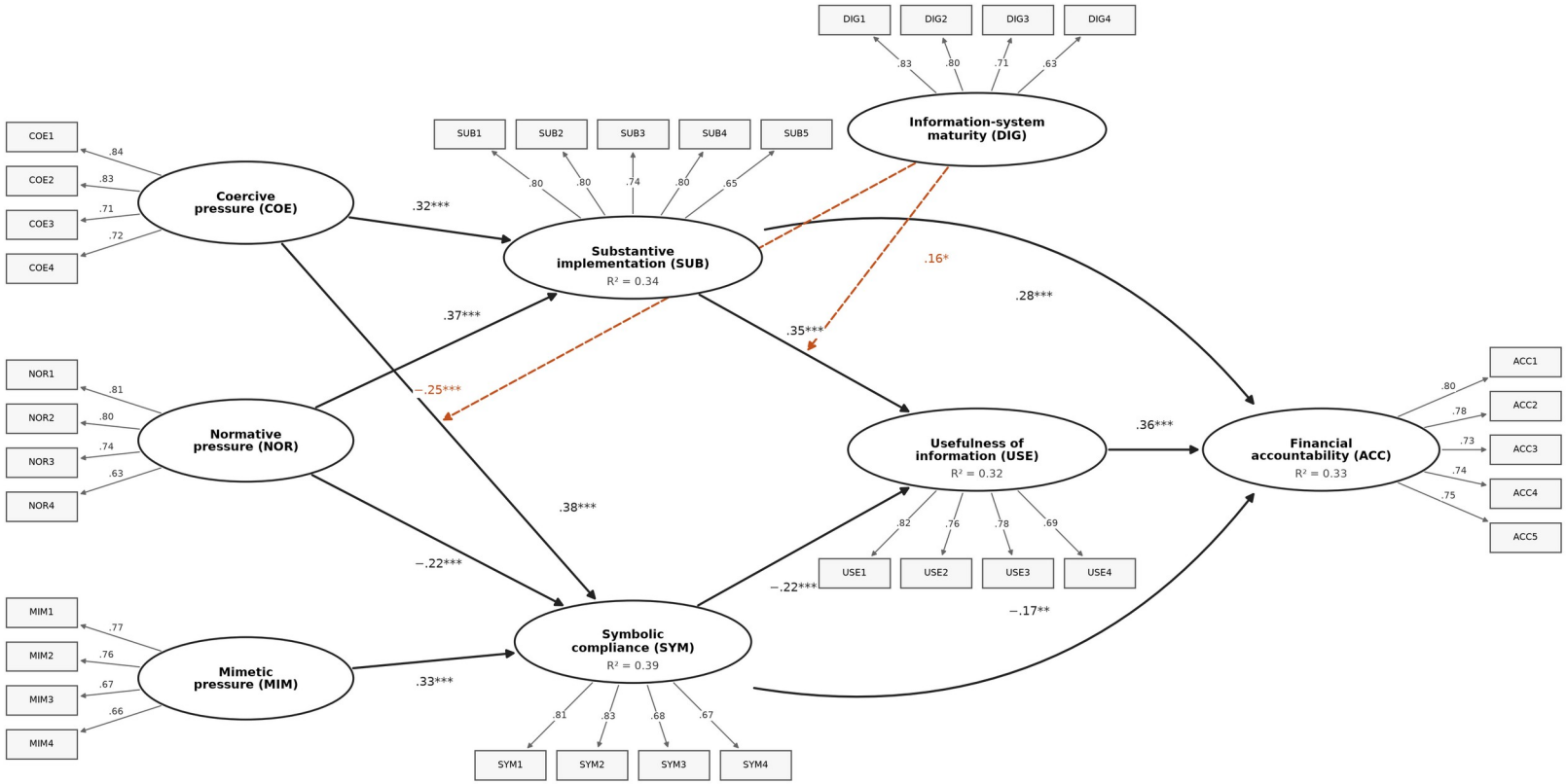
The eight-factor confirmatory model fitted well: $\chi^2(499) = 548.1$, $p = .063$; CFI = 0.990, TLI = 0.989, RMSEA = 0.017, SRMR = 0.040. Standardized loadings ranged from 0.63 to 0.84 (Figure 3, Appendix A), composite reliability from 0.81 to 0.87 and AVE from 0.51 to 0.60 (Table 2), all above conventional thresholds although not at the ceiling that would suggest redundant items. The highest HTMT was 0.52, far below .85, and the square root of each AVE exceeded the corresponding inter-construct correlations (Appendix B). Harman's first factor explained 20.5% of variance; a one-factor CFA fitted poorly (CFI = 0.33) and adding a common latent factor changed fit marginally (CFI = 0.999; common variance 20%), so method variance is unlikely to drive the results. Latent variance inflation factors in all equations were below 1.30.

Table 2. Reliability and convergent validity

Construct	Items	α	CR	AVE	Loadings
Coercive pressure (COE)	4	0.86	0.86	0.60	0.71–0.84
Normative pressure (NOR)	4	0.83	0.83	0.56	0.62–0.81
Mimetic pressure (MIM)	4	0.80	0.81	0.51	0.66–0.77
Substantive implementation (SUB)	5	0.87	0.87	0.58	0.65–0.81
Symbolic compliance (SYM)	4	0.83	0.84	0.56	0.67–0.84
Information-system maturity (DIG)	4	0.83	0.83	0.55	0.63–0.83
Usefulness of reported information (USE)	4	0.85	0.85	0.58	0.69–0.82
Financial accountability (ACC)	5	0.87	0.87	0.58	0.73–0.80

4.3 Structural model

The full structural model fitted well: $\chi^2(830) = 921.0$, $p = .015$; CFI = 0.984, TLI = 0.982, RMSEA = 0.018, SRMR = 0.043. Figure 2 shows the standardized path coefficients and Table 3 reports unstandardized and standardized estimates with the verdict on each hypothesis.



Standardized estimates from the full structural model (maximum likelihood). Arrows to boxes: standardized loadings (λ); arrows between latent variables: standardized path coefficients (β).
* p < .05, ** p < .01, *** p < .001. Dashed orange: latent interaction paths (maturity × substantive implementation; maturity × coercive pressure), product indicators omitted.
Non-hypothesised paths (mimetic → substantive; maturity main effects) and the control (experience) are estimated but not drawn.

Figure 2. Estimated structural model with all 34 indicators: standardized loadings (λ) and standardized path coefficients (β), with latent R^2. Latent variables are ellipses, indicators are boxes; dashed orange arrows are latent interaction (moderation) terms.

Table 3. Structural estimates

Hyp.	Path	b	SE	β	p	Verdict
H1	Coercive → Substantive implementation	0.31	0.06	0.32	< .001	Supported
H2	Coercive → Symbolic compliance	0.38	0.06	0.38	< .001	Supported
H3	Normative → Substantive implementation	0.38	0.06	0.37	< .001	Supported
H4	Normative → Symbolic compliance	−0.23	0.07	−0.22	< .001	Supported
H5	Mimetic → Symbolic compliance	0.36	0.07	0.33	< .001	Supported
H6	Substantive impl. → Usefulness	0.36	0.06	0.35	< .001	Supported
H7	Symbolic compliance → Usefulness	−0.22	0.06	−0.22	< .001	Supported
H8	Substantive impl. → Accountability	0.27	0.06	0.28	< .001	Supported
H9	Symbolic compliance → Accountability	−0.16	0.06	−0.17	.005	Supported
H10	Usefulness → Accountability	0.35	0.07	0.36	< .001	Supported
H11	Maturity × Subst. impl. → Usefulness	0.18	0.07	0.16	.013	Supported
H12	Maturity × Coercive → Symbolic compliance	−0.30	0.08	−0.25	< .001	Supported
–	Mimetic → Substantive implementation	0.08	0.07	0.07	.245	–
–	Maturity → Substantive implementation	0.04	0.06	0.03	.542	–
–	Maturity → Symbolic compliance	−0.29	0.06	−0.27	< .001	–
–	Maturity → Usefulness	0.30	0.06	0.28	< .001	–
–	Experience → Accountability (control)	0.03	0.04	0.04	.442	–

Pressures and responses. Coercive pressure raised both substantive implementation (β = 0.32, p < .001; H1) and symbolic compliance (β = 0.38, p < .001; H2). Normative pressure was the strongest predictor of substantive implementation (β = 0.37, p < .001; H3) and reduced symbolic compliance (β = −0.22, p < .001; H4). Mimetic pressure raised symbolic compliance (β = 0.33, p < .001; H5) and had no reliable effect on substantive implementation (β = 0.07, p = .245). The pattern is the signature of three different mechanisms: mandates produce activity of both kinds; professional norms push towards substance; templates push towards form.

Responses and outcomes. Substantive implementation raised the usefulness of information (β = 0.35, p < .001; H6) and accountability (β = 0.28, p < .001; H8), whereas symbolic compliance lowered usefulness (β = −0.22, p < .001; H7) and accountability (β = −0.17, p = .005; H9). Usefulness was in turn positively related to accountability (β = 0.36, p < .001; H10), which identifies it as a mechanism that carries part of the effect of both responses. Explained variance (latent R^2) was 0.34 for substantive implementation, 0.39 for symbolic compliance, 0.32 for usefulness and 0.33 for accountability, which in cross-sectional organisational research indicates substantial explanatory power.

Moderation. Information-system maturity strengthened the effect of substantive implementation on usefulness (β = 0.16, p = .013; H11) and weakened the effect of coercive pressure on symbolic compliance (β = −0.25, p < .001; H12). Effect sizes were small to moderate (f^2 = 0.035 for usefulness and 0.103 for symbolic compliance), as is typical for interaction effects in non-experimental data. Figure 4 plots the conditional slopes: at low maturity the slope of usefulness on implementation is roughly 0.21, at high maturity 0.51; the slope of symbolic compliance on coercion falls from 0.63 to 0.12.

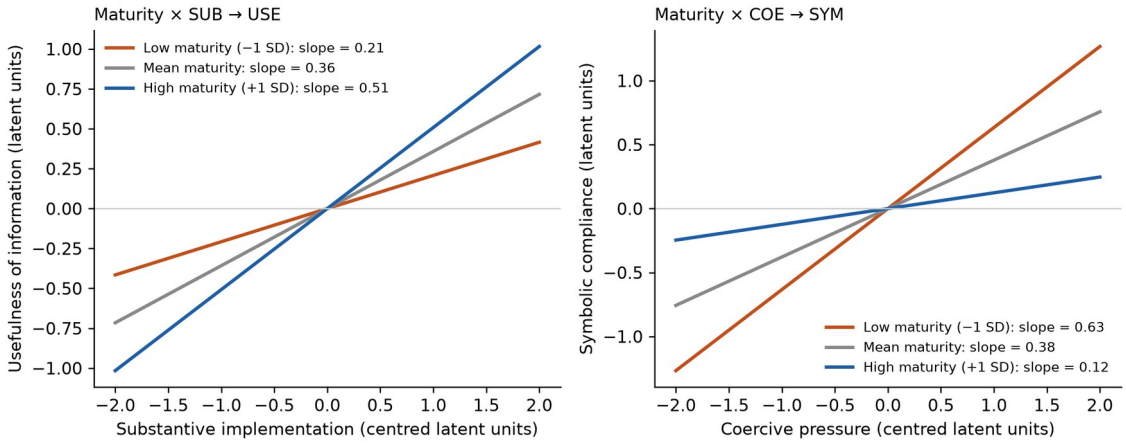


Figure 4. Conditional effects (simple slopes) of the two latent interactions at −1 SD, the mean and +1 SD of information-system maturity.

4.4 Indirect and conditional indirect effects

Table 4. Conditional indirect effects of institutional pressures on financial accountability (unstandardized; 95% bootstrap CI)

Route	Maturity −1 SD	Mean	+1 SD
Coercive pressure via substantive implementation	0.11, 95% CI [0.05, 0.17]	0.12, 95% CI [0.07, 0.19]	0.14, 95% CI [0.08, 0.22]
Coercive pressure via symbolic compliance	−0.15, 95% CI [−0.24, −0.08]	−0.09, 95% CI [−0.14, −0.05]	−0.03, 95% CI [−0.07, 0.01]
Coercive pressure: total indirect effect	−0.04, 95% CI [−0.15, 0.05]	0.03, 95% CI [−0.04, 0.11]	0.11, 95% CI [0.04, 0.20]
Normative pressure via substantive implementation	0.13, 95% CI [0.07, 0.21]	0.15, 95% CI [0.09, 0.23]	0.17, 95% CI [0.11, 0.25]
Normative pressure via symbolic compliance	0.06, 95% CI [0.02, 0.10]	0.06, 95% CI [0.02, 0.10]	0.06, 95% CI [0.02, 0.10]
Normative pressure: total indirect effect	0.19, 95% CI [0.11, 0.28]	0.21, 95% CI [0.13, 0.30]	0.23, 95% CI [0.15, 0.33]
Mimetic pressure via substantive implementation	0.03, 95% CI [−0.02, 0.08]	0.03, 95% CI [−0.02, 0.08]	0.03, 95% CI [−0.02, 0.09]
Mimetic pressure via symbolic compliance	−0.09, 95% CI [−0.15, −0.04]	−0.09, 95% CI [−0.15, −0.04]	−0.09, 95% CI [−0.15, −0.04]
Mimetic pressure: total indirect effect	−0.06, 95% CI [−0.13, 0.01]	−0.05, 95% CI [−0.14, 0.02]	−0.05, 95% CI [−0.14, 0.03]

At mean maturity, coercive pressure had a positive indirect effect on accountability through substantive implementation (0.12, 95% CI [0.07, 0.19]) and a negative one through symbolic compliance (−0.09, 95% CI [−0.14, −0.05]); the two almost cancel, leaving a total indirect effect of 0.03, 95% CI [−0.04, 0.11]. Normative pressure had a clearly positive total indirect effect (0.21, 95% CI [0.13, 0.30]), and mimetic pressure a negative effect through symbolic compliance (−0.09, 95% CI [−0.15, −0.04]) with a total that is statistically indistinguishable from zero (−0.05, 95% CI [−0.14, 0.02]). H13 is therefore supported. Maturity changes the arithmetic of coercion: the total indirect effect of coercive pressure moves from −0.04, 95% CI [−0.15, 0.05] at low maturity to 0.11, 95% CI [0.04, 0.20] at high maturity, because both the ceremonial route weakens and the substantive route is amplified (Figure 5). Indices of moderated mediation were 0.061, 95% CI [0.024, 0.116] for the

coercion–ceremony route and 0.052, 95% CI [0.007, 0.117] for the implementation–usefulness–accountability route.

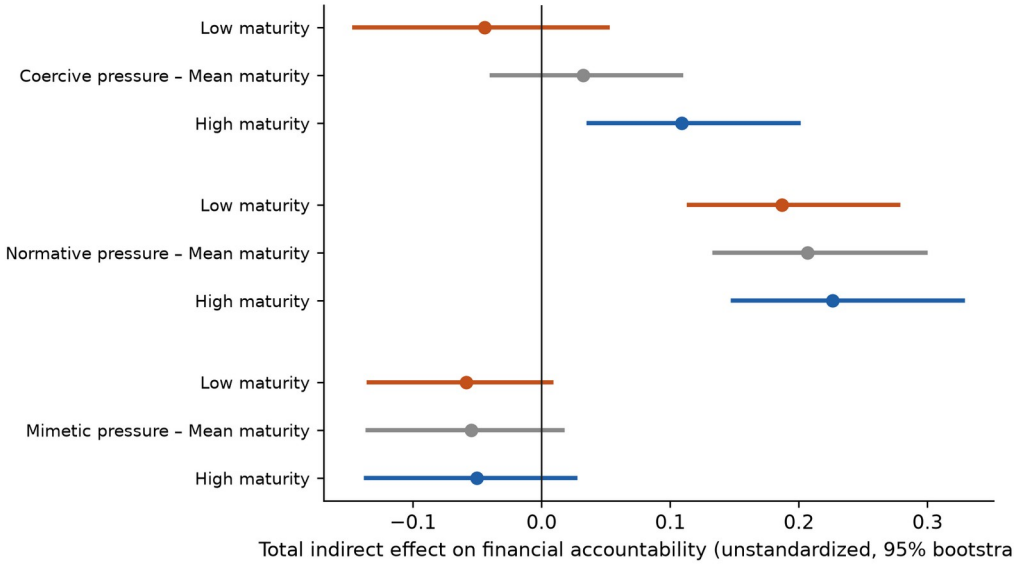


Figure 5. Total indirect effects of the three pressures on financial accountability at three levels of information-system maturity (95% bootstrap CIs).

4.5 Competing models

Table 5. Competing structural models

Model	Specification	χ^2	df	CFI	RMSEA	AIC	BIC
M1	Hypothesised model	921.0	830	0.984	0.018	227	669
M2	No moderation	944.2	832	0.980	0.020	222	658
M3	No usefulness path	950.2	831	0.979	0.021	224	663
M4	Substantive route only	943.4	831	0.980	0.020	224	663
M5	Direct effects of pressures added	918.5	827	0.984	0.018	233	687

The hypothesised model fitted better than the model without moderation (M2; $\Delta\chi^2(2) = 23.2$, $p < .001$), the model without the usefulness path (M3; $\Delta\chi^2(1) = 29.2$, $p < .001$) and the substantive-route-only model (M4; $\Delta\chi^2(1) = 22.4$, $p < .001$). Adding direct paths from the three pressures to accountability (M5) did not improve fit ($\Delta\chi^2(3) = 2.5$, n.s.), consistent with full mediation. BIC values from the software favour the more parsimonious structures; given the large number of product-indicator parameters we rely on the nested χ^2 tests and the fit indices for model selection.

4.6 Robustness

Table 6. Robustness of standardized paths across estimators and subsamples

Path	Baseline	DWLS	Residual-centred	Respecified	Trimmed 5%	Experienced (≥10 yrs)
Coercive → Substantive	0.32	0.33	0.32	0.32	0.32	0.28
Coercive → Symbolic	0.38	0.32	0.38	0.38	0.39	0.43
Normative → Substantive	0.37	0.38	0.37	0.37	0.37	0.38
Normative → Symbolic	−0.22	−0.22	−0.22	−0.22	−0.23	−0.24

Path	Baseline	DWLS	Residual-centred	Respecified	Trimmed 5%	Experienced (≥10 yrs)
Mimetic → Symbolic	0.33	0.32	0.33	0.33	0.33	0.32
Substantive → Usefulness	0.35	0.38	0.36	0.35	0.37	0.31
Symbolic → Usefulness	−0.22	−0.23	−0.22	−0.22	−0.24	−0.22
Usefulness → Accountability	0.36	0.35	0.36	0.36	0.41	0.39

Estimates were stable in sign and magnitude when estimated by DWLS without interaction terms (CFI = 1.018), with residual-centred product indicators (CFI = 0.997), after adding the four largest within-construct residual correlations (ACC4–ACC5, ACC2–ACC4, ACC1–ACC2, DIG3–DIG4; CFI = 0.987), after removing the 17 most extreme multivariate observations, and in the subsample of 233 respondents with at least ten years of experience (Table 6; Appendix E).

5. Discussion

5.1 What the results say

The central result is that the same institutional pressure can feed opposite organisational behaviours. Coercion is a blunt instrument: it raises substantive implementation and symbolic compliance in comparable measure, so that the net effect on accountability is close to zero unless something else tips the balance. Normative pressure, in contrast, works almost entirely through substance, and mimetic pressure almost entirely through form. These contrasts support the view that "adoption" is an insufficient dependent variable; what matters is the quality of adoption.

Second, the usefulness of information acts as the link between implementation and accountability. When implementation is substantive, information is in use, so it is timely and trusted; when compliance is symbolic, information is discounted and accountability is weaker. This extends the literature on performance-information use to accrual and budgeting reforms and indicates that accountability gains require that the information be used for decisions, not merely produced.

Third, information-system maturity is both an amplifier and a safeguard. It amplifies the payoff of substantive implementation, as integrated and real-time data make cost and accrual information usable. It also safeguards against ceremony: as systems become integrated and auditable, the coercion–ceremony pathway weakens, because mandated reporting leaves traces that can be verified. The finding that maturity changes the sign pattern of the total indirect effect of coercion from near-zero or negative at low maturity to positive at high maturity is, in our view, the most policy-relevant result: mandates are productive when the infrastructure exists to make them verifiable.

5.2 Theoretical implications

For institutional theory in accounting (Modell, 2009; Brignall & Modell, 2000), the study offers an empirical specification of the distinction between ceremonial adoption and internalisation (Kostova & Roth, 2002) and shows that the three isomorphic mechanisms map onto this distinction differently. It supports arguments that normative professionalisation can counter decoupling (Bromley & Powell, 2012; Greenwood et al., 2011) and identifies a

boundary condition—system maturity—that is rarely theorised in institutional accounts. For the accountability literature (Bovens, 2007; Bracci et al., 2015) the results suggest that information is not a sufficient condition for accountability; its effect depends on whether it is used, which is itself a function of the organisation's response to pressure.

5.3 Implications for practice and policy

Do not rely on mandates alone. Because coercion also stimulates ceremony, legal requirements should be complemented by professional norms: certification, continuing education, peer learning networks and visible leadership by professional bodies.

Monitor use, not submission. Oversight bodies can reduce ceremonial incentives by auditing the use of cost and accrual information in allocation decisions (for example, whether unit cost informed a budget request) rather than the timeliness or completeness of reports.

Invest in infrastructure as part of reform design. Integrated, real-time and auditable systems raise the payoff of substantive implementation and make ceremony costly; they should be planned with, not after, accounting reforms.

Localise templates. Where agencies copy templates from peers or consultants, the model predicts ceremonial results; standard formats should be accompanied by guidance and capacity-building to adapt them.

Design indicators for use. Performance indicators that are easy to meet signal ceremony; benchmarking against history and external standards can reduce this tendency.

6. Limitations and future research

The design is cross-sectional and based on self-reports from one respondent per agency, so causal ordering cannot be established and common-source bias, although not indicated by our tests, cannot be ruled out; matched respondent pairs, objective indicators (budget variance, performance-audit findings) and longitudinal data would strengthen inference. The scales for substantive implementation, symbolic compliance and accountability are new and need replication, cross-group invariance testing and, ideally, validation against behavioural criteria. The sample covers one country with a particular institutional configuration; comparative studies could examine whether the pressure–response mapping depends on the legal system and the development of the profession. Finally, we model two responses and one boundary condition; future work could add leadership, organisational capacity and political-administrative relations, and could test configurations of pressures (for example, synergies between coercive and normative pressures), which we examine in a companion study.

7. Conclusion

Financial reforms succeed less because they are required than because of how agencies respond to the requirement. The evidence presented here shows that coercion can yield form as readily as substance, that professional norms favour substance, that imitation favours form, and that integrated, auditable information systems both amplify the benefits of substantive implementation and limit the pull of ceremony. Reformers who want accountability rather than compliance should invest in norms, use-focused oversight and infrastructure, and not in mandates alone.

Declarations

Funding: [to be completed]. **Competing interests:** [to be completed]. **Ethics:** Informed consent and ethics approval details of the survey must be stated here. **Data and code availability:** The questionnaire, the analysis script (Python, semopy) and a Colab notebook accompany this manuscript; the dataset is available from the authors on reasonable request. **Acknowledgements:** [to be completed].

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Appendix A. Instrument, origin of each scale and standardized loadings

Response format: 1 = strongly disagree … 5 = strongly agree for all items. Loadings (λ) are standardized estimates in the full structural model. The Persian original of each item, scoring notes and the validation protocol are provided in the accompanying questionnaire file.

Coercive pressure (COE) – origin: Adapted; sources: DiMaggio & Powell (1983); Teo et al. (2003)

Code	Item	λ
COE1	Higher-level laws and regulations oblige our agency to implement accrual accounting and performance-based budgeting.	0.84
COE2	Oversight bodies and higher authorities closely supervise the implementation of these reforms.	0.83
COE3	Failing to implement these reforms has legal or administrative consequences for the agency.	0.71
COE4	Issued directives and circulars specify precisely how the reforms must be implemented.	0.72

Normative pressure (NOR) – origin: Adapted; sources: DiMaggio & Powell (1983); Teo et al. (2003)

Code	Item	λ
NOR1	The public-sector accounting and auditing profession regards implementing accrual accounting as a professional norm.	0.81
NOR2	Training courses and professional gatherings promote the implementation of these reforms.	0.80
NOR3	Experienced colleagues and comptrollers consider careful implementation a professional value.	0.74
NOR4	In academic and professional forums, compliance with public-sector accounting standards is considered desirable.	0.63

Mimetic pressure (MIM) – origin: Adapted; sources: DiMaggio & Powell (1983); Oliver (1991); Teo et al. (2003)

Code	Item	λ
MIM1	Successful similar agencies have implemented the reforms and we follow their example.	0.77
MIM2	When unsure how to implement, we take the method of peer agencies as our model.	0.76
MIM3	We mostly use ready-made formats and reports from consultants or common software.	0.67
MIM4	We worry that if we do not act like other agencies we will be seen as lagging.	0.66

Substantive implementation (SUB) – origin: Researcher-made; sources: Pollitt & Bouckaert (2017); Moynihan (2008); Christiaens et al. (2015)

Code	Item	λ
SUB1	The agency's budget figures are prepared on the basis of programmes and performance indicators.	0.80
SUB2	The agency calculates the full cost of activities and services.	0.80
SUB3	Accrual financial statements are prepared regularly with a defined control process.	0.74
SUB4	Financial and performance reports are reviewed in management meetings and corrective action follows.	0.80
SUB5	Financial staff and programme managers understand the concepts of these reforms and apply them in daily work.	0.65

Symbolic compliance (SYM) – origin: Researcher-made; sources: Meyer & Rowan (1977); Bromley & Powell (2012); Modell (2009)

Code	Item	λ
SYM1	Most performance-budget reports are prepared mainly for submission to higher authorities.	0.81

Code	Item	λ
SYM2	Performance indicators are set so that they are easily achieved.	0.83
SYM3	In practice, the formal implementation of these reforms has no effect on how resources are allocated and spent.	0.68
SYM4	Accrual information is mostly prepared to complete forms and meet requirements rather than for management.	0.67

Information-system maturity (DIG) – origin: Adapted + new; sources: DeLone & McLean (2003); Dunleavy et al. (2006)

Code	Item	λ
DIG1	The agency's financial systems are integrated and data are transferred without re-entry.	0.83
DIG2	Financial and performance information is available to managers online.	0.80
DIG3	The systems preserve a trail of entries and changes (auditability).	0.71
DIG4	Staff have received the training needed to use the systems effectively.	0.63

Usefulness of reported information (USE) – origin: Adapted + new; sources: Moynihan & Pandey (2010); Kroll (2015); DeLone & McLean (2003)

Code	Item	λ
USE1	Reported information is the basis of managerial decisions (allocation and prioritisation) in the agency.	0.82
USE2	Reports are timely and understandable.	0.76
USE3	Accrual information gives a fuller picture of the agency's financial position than the cash basis.	0.78
USE4	Reported information is reliable for evaluating the efficiency and effectiveness of programmes.	0.69

Financial accountability (ACC) – origin: Researcher-made; sources: Bovens (2007); Bracci et al. (2015); Moore (1995)

Code	Item	λ
ACC1	The agency provides or publishes financial and performance information for stakeholders (legislature, oversight bodies, citizens).	0.80
ACC2	Managers explain results and deviations to oversight bodies.	0.78
ACC3	When the budget is exceeded, responsibility is assigned and corrective action is pursued.	0.73
ACC4	Audit and inspection findings are followed up and resolved within a reasonable time.	0.74
ACC5	Information provided by the agency can be verified.	0.75

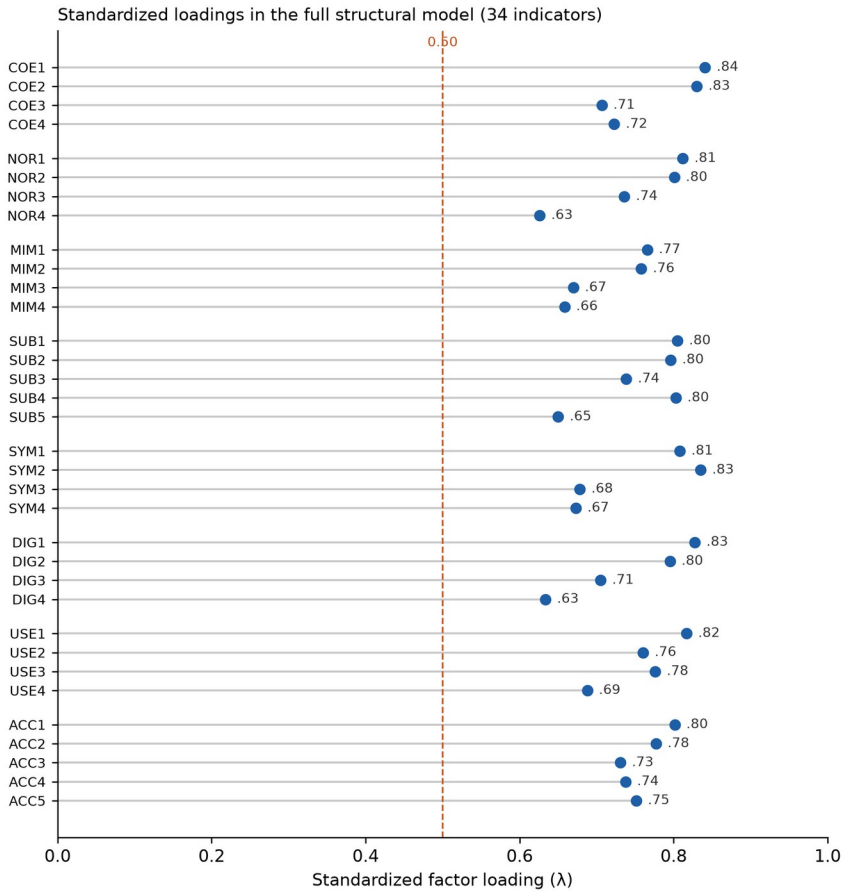


Figure A1. Standardized factor loadings of the 34 indicators in the full structural model.

Appendix B. Latent correlations and discriminant validity

Table B1. Latent correlations (CFA; diagonal = √AVE)

	COE	NOR	MIM	SUB	SYM	DIG	USE	ACC
COE	0.78	0.22	0.32	0.43	0.42	0.12	0.12	0.15
NOR	0.22	0.75	0.33	0.47	−0.02	0.07	0.17	0.20
MIM	0.32	0.33	0.71	0.31	0.35	0.08	0.11	0.11
SUB	0.43	0.47	0.31	0.76	0.03	0.10	0.38	0.40
SYM	0.42	−0.02	0.35	0.03	0.75	−0.22	−0.28	−0.26
DIG	0.12	0.07	0.08	0.10	−0.22	0.74	0.37	0.19
USE	0.12	0.17	0.11	0.38	−0.28	0.37	0.76	0.51
ACC	0.15	0.20	0.11	0.40	−0.26	0.19	0.51	0.76

Table B2. Heterotrait–monotrait ratios (HTMT)

	COE	NOR	MIM	SUB	SYM	DIG	USE	ACC
COE	–	0.22	0.34	0.44	0.40	0.13	0.14	0.16
NOR	0.22	–	0.33	0.47	0.05	0.09	0.18	0.20
MIM	0.34	0.33	–	0.31	0.36	0.10	0.12	0.11
SUB	0.44	0.47	0.31	–	0.06	0.13	0.38	0.40
SYM	0.40	0.05	0.36	0.06	–	0.24	0.28	0.28
DIG	0.13	0.09	0.10	0.13	0.24	–	0.37	0.19
USE	0.14	0.18	0.12	0.38	0.28	0.37	–	0.52
ACC	0.16	0.20	0.11	0.40	0.28	0.19	0.52	–

Appendix C. Complete structural estimates with confidence intervals

Table C1. Unstandardized estimates (Wald 95% CI) and standardized coefficients

Path (outcome ← predictor)	b	SE	95% CI	β
SUB ← COE	0.307	0.058	[0.193, 0.422]	0.320
SUB ← NOR	0.381	0.065	[0.254, 0.507]	0.370
SUB ← MIM	0.077	0.066	[−0.053, 0.208]	0.073
SUB ← DIG	0.035	0.058	[−0.078, 0.149]	0.033
SYM ← COE	0.379	0.062	[0.257, 0.500]	0.384
SYM ← NOR	−0.234	0.066	[−0.363, −0.105]	−0.222
SYM ← MIM	0.359	0.074	[0.214, 0.503]	0.332
SYM ← DIG	−0.294	0.063	[−0.417, −0.171]	−0.272
SYM ← INTC	−0.304	0.079	[−0.458, −0.149]	−0.249
USE ← SUB	0.358	0.061	[0.239, 0.477]	0.352
USE ← SYM	−0.217	0.060	[−0.334, −0.100]	−0.219
USE ← DIG	0.304	0.065	[0.177, 0.432]	0.284
USE ← INTS	0.178	0.072	[0.038, 0.319]	0.156
ACC ← SUB	0.273	0.062	[0.151, 0.395]	0.278
ACC ← SYM	−0.162	0.058	[−0.275, −0.049]	−0.170
ACC ← USE	0.346	0.065	[0.219, 0.474]	0.359
ACC ← exp	0.034	0.044	[−0.052, 0.120]	0.039

Appendix D. Moderated-mediation indices and conditional effects

Table D1. Indices of moderated mediation, conditional effects of substantive implementation and standardized total indirect effects

Quantity	Estimate	95% bootstrap CI	CI excludes 0
SUB_USE_ACC_low	0.072	[0.004, 0.150]	Yes
SUB_all_low	0.345	[0.209, 0.483]	Yes
SUB_USE_ACC_mean	0.124	[0.073, 0.192]	Yes
SUB_all_mean	0.397	[0.275, 0.525]	Yes
SUB_USE_ACC_high	0.176	[0.098, 0.278]	Yes
SUB_all_high	0.449	[0.318, 0.589]	Yes
imm_COE_SYM	0.061	[0.024, 0.116]	Yes
imm_COE_SUB	0.016	[0.002, 0.039]	Yes
imm_NOR_SUB	0.020	[0.003, 0.045]	Yes
imm_MIM_SUB	0.004	[−0.003, 0.014]	No
imm_SUB_ACC	0.052	[0.007, 0.117]	Yes
std_COE_tot	0.034	[−0.043, 0.114]	No
std_NOR_tot	0.205	[0.137, 0.281]	Yes
std_MIM_tot	−0.053	[−0.126, 0.016]	No

Appendix E. Measurement-model checks and full robustness fit

Table E1. Fit statistics across specifications

Specification	χ^2	df	CFI	TLI	RMSEA	SRMR
Eight-factor CFA	548.1	499	0.990	0.989	0.017	0.040
One-factor CFA	3921.9	527	0.330	0.287	0.139	0.153
CFA + common latent factor	471.4	465	0.999	0.998	0.006	0.030

Specification	χ^2	df	CFI	TLI	RMSEA	SRMR
Structural model (baseline)	921.0	830	0.984	0.982	0.018	0.043
DWLS, no interactions	363.7	541	1.018	1.020	0.000	0.042
Residual-centred indicators	847.7	830	0.997	0.997	0.008	0.042
Respecified (4 residual correlations)	896.9	826	0.987	0.986	0.016	0.043
Trimmed 5%	943.3	830	0.979	0.978	0.021	0.045
Experienced subsample	972.3	830	0.965	0.961	0.027	0.051